

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			award (1) each for any two of following - used equal volumes/amounts of each water sample - shook each water sample for an equal amount of time - used the same concentration of soap solution for each water sample	2			2		2
	(b)	(i)		$\frac{18 + 19 + 17}{3}$ (1) must show working 18 (1) award (1) max for 18 with no working		2		2	2	
		(ii)		A (1) needed highest volume/amount of soap solution (to form permanent lather) (1) ecf possible from part (i)			2	2		2
	(c)			all three of following for (2) award (1) for any one/two - wastes soap - forms a scum with soap - forms limescale when heated	2			2		
	(d)			Ca ²⁺	1			1		
				Question 2 total	5	2	2	9	2	4